

FinDocAgent: A Multimodal AI Agent for Financial Document Understanding and Risk Analysis

Tinara Witharana

ID: 250166619

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Supervisor: Dr. Ammar Yasir Naich

1. Introduction & Motivation

Financial institutions process thousands of documents daily such as bank statements, KYC identity forms, invoices, and earnings reports where each contains a mixture of printed text, tables, signatures, stamps, and handwritten annotations. Automating this process is a multi-billion dollar problem in compliance, fraud detection, and risk management. Existing AI systems either treat documents as pure text, losing all visual layout information, or classify images without reasoning over their content. Neither approach can replicate what an analyst does: read an invoice, notice the line items do not sum to the stated total, and produce a written explanation flagging the discrepancy.

FinDocAgent is a multimodal AI agent that accepts a financial document as input and autonomously extracts structured data, detects anomalies, and produces an explainable risk report, combining state-of-the-art vision-language models with agentic reasoning frameworks.

2. Problem Statement & Research Gap

Visual document models such as LayoutLMv3 Huang et al. [1] and Qwen2-VL Wang et al. [2] represent the state-of-the-art in perceiving document content, with Qwen2-VL approaching human-level performance on the DocVQA benchmark [3]. However, both are fundamentally static and single-pass, which means they answer one question at a time, hold no memory across queries, and cannot call external tools or update their reasoning based on partial results.

Agentic frameworks such as ReAct Yao et al. [4] and Toolformer Schick et al. [5] address exactly those limitations which enables LLMs to reason iteratively, call tools, and update decisions dynamically. However, both operate entirely on text and are blind to the visual structure of documents such as tables, layout, stamps, and form fields that carry critical information in financial documents.

The gap is that there is no existing system that combines multimodal visual document parsing, multi-step agentic tool orchestration, and explainable anomaly detection in the financial domain. This is not only a technical gap, the EU AI Act (2024) and FCA guidelines both require automated financial decisions to be auditable and explainable, meaning a system that cannot justify its outputs is not deployable in practice regardless of accuracy [6].

This project addresses that gap through three research questions:

1. **RQ1:** Does agentic multi-step reasoning outperform single-pass baselines on financial document understanding, measured by ANLS and F1?
2. **RQ2:** Does multimodal visual encoding improve extraction accuracy over text-only pipelines on documents containing tables and form fields?
3. **RQ3:** Can attention maps and SHAP values provide document-region-level explanations sufficient for financial compliance audit requirements?

The primary contribution of this project is FinDocAgent. An end-to-end system combining Qwen2-VL multimodal encoding, LangGraph agentic orchestration, and an XAI explainability layer which is

evaluated on DocVQA, FinTabNet, and real SEC 10-K filings.

3. Proposed System Architecture

FinDocAgent is composed of five integrated layers, illustrated in Figure 1.

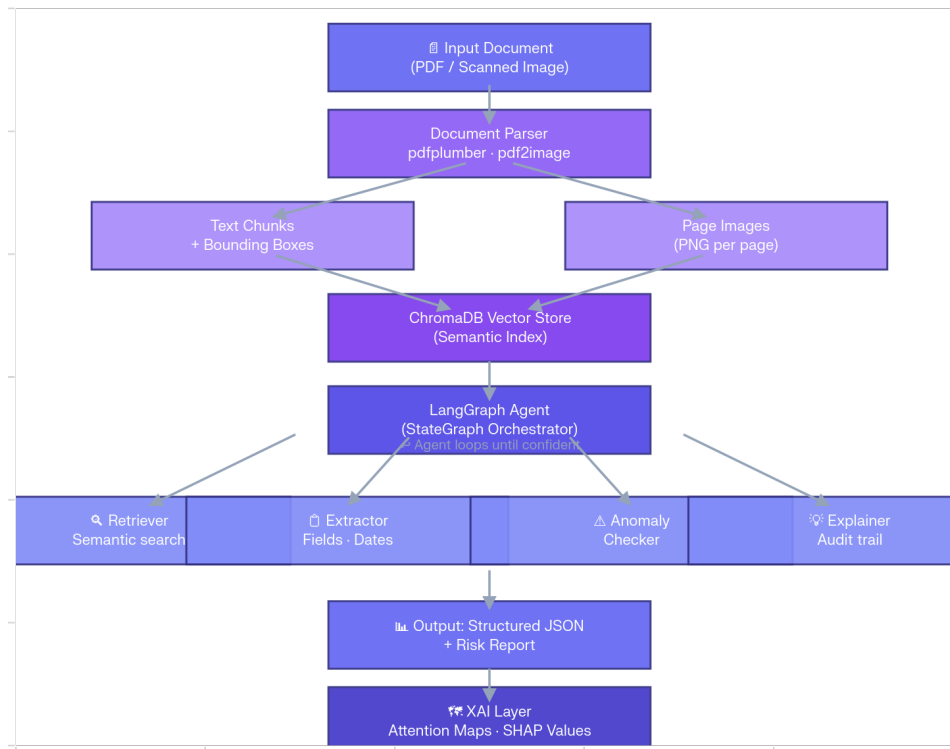


Figure 1. FinDocAgent end-to-end system architecture.

Layer 1 (Document Parser): Financial documents come as digital PDFs or scanned images. Using `pdfplumber` and `pdf2image`, the parser extracts raw text with spatial bounding box coordinates (the precise position of every word on the page) and converts each page to a high-resolution image. The spatial coordinates are critical to know whether a number sits in the “Total” row or the “Subtotal” row.

Layer 2 (Vision Encoder - Qwen2-VL): Page images are processed by Qwen2-VL [2], a vision-language model pre-trained on billions of image-text pairs. It interprets visual elements that text parsers cannot such as tables, stamps, form fields, and handwritten text and then produces a rich vector embeddings that encode both visual appearance and semantic meaning.

Layer 3 (ChromaDB Vector Store): All document chunks, text and visual embeddings, are indexed in ChromaDB. Rather than loading an entire 200-page SEC filing into the model’s context window on every query, ChromaDB performs semantic search which retrieves only the most relevant chunks by meaning, not keyword matching. This makes the system scalable to real-world long financial documents.

Layer 4 (LangGraph Agent Orchestrator): The LangGraph StateGraph is the reasoning core. It maintains a running memory of extracted fields, completed checks, and prior decisions, autonomously routing between four tools and looping back when confidence is low:

- Retriever — fetches semantically relevant document chunks
- Extractor — pulls structured fields such as names, dates, amounts
- Anomaly Checker — cross-validates fields against business rules (e.g., do line item totals match the invoice total?)
- Explainer — generates a human-readable audit trail citing specific document regions.

Layer 5 (XAI Layer): The output is accompanied by attention map highlighting which document regions drove each decision, and SHAP values quantifying each field’s contribution to the anomaly score. This makes FinDocAgent deployable in regulated environments where black-box AI decisions are unacceptable.

4. Evaluation Plan

The following experiments will compare the agentic multimodal approach against simpler baselines. The primary metric is ANLS (Answer Normalised Levenshtein Similarity), the standard benchmark metric for document VQA.

Table 1. Planned ablation experiments and evaluation metrics.

Experiment	What It Tests	Metric
Single-pass VLM vs. FinDocAgent	Agentic reasoning improves accuracy	ANLS, F1
Text-only vs. Multimodal pipeline	Vision encoding is necessary	ANLS on DocVQA
Single-tool vs. Multi-tool agent	Tool routing adds value	Extraction accuracy
Without XAI vs. With XAI	Explainability does not hurt performance	Human evaluation
FinDocAgent vs. GPT-4o baseline	Competitiveness of open-source approach	ANLS, latency

Datasets: DocVQA [3] (50,000 questions over 12,767 document images); FinTabNet (financial table extraction); FUNSD (noisy form understanding); SEC 10-K Filings (real-world public financial documents).

5. Project Plan

The project spans 5 months across six phases, shown in Figure 2.

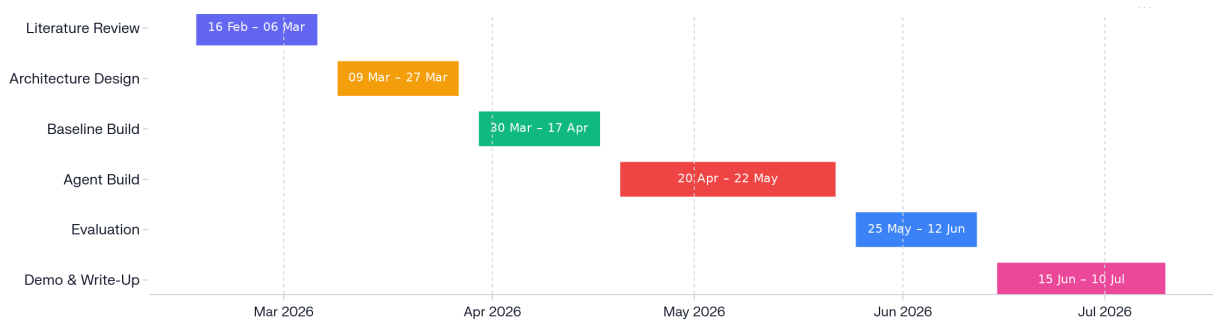


Figure 2. Project Timeline.

Phase 1 — Literature Review: A structured review of the core literature across three areas: visual document understanding, agentic AI frameworks, and financial document AI. Key papers including LayoutLMv3, Qwen2-VL, ReAct, Toolformer are important papers that uses this architecture. By the end of this phase, the research gap will be formally defined and the central research questions confirmed.

Phase 2 — Architecture Design: The full system architecture will be designed before any code is written. This includes defining the data flow between all five layers, specifying the LangGraph agent state schema, designing the four tool interfaces (Retriever, Extractor, Anomaly Checker, Explainer), and selecting the final technology stack. The output of this phase is a complete architecture diagram and a component specification document.

Phase 3 — Baseline Build: A simple, non-agentic single-pass pipeline will be implemented first using Qwen2-VL and pdfplumber. This baseline system answers document queries in one pass with no memory or tool routing. It is evaluated on DocVQA and FinTabNet to produce benchmark ANLS and F1 scores, which will serve as the comparison point for all subsequent experiments.

Phase 4 — Agent Build: This is the core development phase. The LangGraph StateGraph orchestrator will be implemented and progressively integrated with the vision encoder, ChromaDB vector store, and all four tool nodes. The agent will be tested iteratively and each tool node is validated in isolation before being connected to the full graph. By the end of this phase, FinDocAgent runs end-to-end on all three target document types: invoices, bank statements, and KYC forms.

Phase 5 — Evaluation: All experiments will be conducted systematically - agent versus baseline, multimodal versus text-only, single-tool versus multi-tool, and FinDocAgent versus a GPT-4o baseline. Results will be recorded across ANLS, F1, precision, recall, and latency. The XAI module will be evaluated through a structured human assessment, in which attention map visualisations and SHAP explanations are scored against a predefined set of financial compliance criteria across a sample of 25 test documents.

Phase 6 — Demo & Write-Up: The final phase covers both deployment and academic write-up. A Streamlit demo application will be built and deployed on HuggingFace Spaces, providing a live interface for uploading financial documents and viewing agent outputs. In parallel, the full dissertation will be written, incorporating all evaluation results, architecture descriptions, and literature review. The GitHub repository will be cleaned and documented for open-source release before final submission.

References

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